

Utkarsh Kailas Raut

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GitHub: github.com/utkarsh-raut

Skills

- **Languages:** C, Python
- **Technologies/Tools:** Linux, Alloy
- **Databases:** PostgreSQL

Projects

- **Neural Network for Digit Detection**

Built a neural network from scratch for handwritten digit recognition without using high-level ML libraries. Implemented core mathematical operations manually, achieving 96% accuracy with a single hidden layer. Worked on image preprocessing and model computation using Python.

Technologies: Python, NumPy, Pillow

- **Real-Time Gesture-Controlled Virtual Mouse**

Built a vision-based human-computer interaction system enabling touchless mouse control using hand gestures. Designed gesture recognition logic using hand landmarks for multi-action input including movement, click, drag, and idle states. Improved usability through exponential smoothing, active-zone scaling, and noise reduction techniques for stable cursor behavior.

Technologies: Python, OpenCV, MediaPipe, PyAutoGUI

- **CPU Scheduler (Raspberry Pi Pico - RP2040) (Ongoing)**

Designing and implementing a CPU scheduler on Raspberry Pi Pico, exploring scheduling algorithms such as FCFS and Round Robin. Analyzing execution flow and low-level system behavior in an embedded environment.

Education

Master of Computer Science (Pursuing)

2024 – 2026

Department of Computer Science, Pune University

Bachelor of Computer Application

2021 – 2024

Dr APJ Abdul Kalam College

Additional Skills & Strengths

- Strong problem-solving skills
- Quick learner with the ability to adapt to new technologies